ECSE 522 Nonlinear Phenomena in Power System Dynamics

Course Project

Due on April 17th, 2026

Chenyi Xu 260948311

Problem 1

(a) In obtaining the new Y-bus,

i. The most typical switching events to consider are transmission line or feeder closing, transformer switching, load switching, shunt reactor switching, and generator switching.

ii. For Y-bus modification, depending on the situation, we can add or remove branch admittance to account for line or transformer switching. Diagonal elements are adjusted to account for shunt switching, while both diagonal and off-diagonal elements are modified to reflect changes in direct bus-to-bus connectivity. If we want to reduce the Y bus to the generator internal node only, we can apply Kron Reduction.

(b) Consider the block where the bus powers are computed.

i. N is the number of generators.

ii. The equation used to compute P_{ei} is shown below:

$$P_{ei} = E_i^2 G_{ii} + \sum_{\substack{j=1 \\ j \neq i}}^{n} E_i E_j Y_{ij} \cos(\theta_{ij} - \delta_i + \delta_j)$$

iii. When computing P_{ei} for a particular bus i, we must obtain voltages at other nodes.

A. No, we do not need to obtain the terminal bus voltage. The network has been reduced to the generator's internal nodes. P_{ei} is computed directly using the generator internal voltage, rotor angle, and reduced Y matrix.

B. Based on the equation above, other generator internal nodes j = 1,...,N are the relevant nodes. We need their internal voltage magnitudes, the rotor angles, the admittance matrix and the angle of admittance.

C. For **constant current loads**, the load draws a fixed current magnitude regardless of voltage change. Power changes proportionally to voltage while current stays the same. Since current is fixed, the admittance has to vary to satisfy $I = YV$. Instead of absorb them into the Y-bus as shunt admittance, we can model them as a current injection. So after we apply the reduction, we have a reduced Y bus term and an extra current source term. We need to include both of them to obtain the power equation. For **constant power loads**, the load maintains the same real and reactive power demand. So in case the voltage changes, the current needs to change to keep the same power. The problem is nonlinear. At each time step, we need to perform iterative analysis to compute the load current and voltages. We can use a method such as Newton-Raphson to solve it. Once the load current is computed, it is used in the network equation in the same way as a current injection term. It is updated whenever there is a voltage change.

(c) Consider the integration scheme represented only in this block.

i. It is a Forward Euler integration scheme.

ii. Because P_{ei} depends on the current machine states. After one integration step, the states change. Since the rotor angle changes, the electrical output changes as well. If we don't recalculate, we will use inaccurate network power in the next step.

iii. Because for each of the N generators, there are 2 state variables, rotor angle and rotor speed. Each state variable has one equation, so together we have 2N equations.

iv. Give the form of these equations for 1 generator in terms of angles, speeds, P_{mi} and P_{ei}.

$$\frac{d\delta_i}{dt} = \omega_i - \omega_{rated}, \qquad \frac{d\omega_i}{dt} = \frac{\omega_{rated}}{2H_i}(P_{mi} - P_{ei})$$

(d,e) Consider the steps in blocks d and e.

i. The integration scheme for d and is a predictor-corrector method. A corrected derivative is obtained in d, then the Forward Euler is applied in e.

ii. Give the form of the equations in steps d and e in terms of angles, speeds, P_{mi} and P_{ei}.
For part d:

$$f^c_{\delta i} = \frac{1}{2}\left[(\omega_i(t-T) - \omega_{rated}) + (\omega^p_i(t) - \omega_{rated})\right]$$

$$f^c_{\omega i} = \frac{1}{2}\left[\frac{\omega_{rated}}{2H_i}(P_{mi} - P_{ei}(t-T)) + \frac{\omega_{rated}}{2H_i}(P_{mi} - P^p_{ei}(t))\right]$$

For part e:

$$\delta_i(t) = \delta_i(t-T) + Tf^c_{\delta_i}$$

$$\omega_i(t) = \omega_i(t-T) + Tf^c_{\omega_i}$$

Problem 2

(a) A Matlab program was written. The complete code is attached.

(a) Internal machine voltage magnitude and angles

```
E_mag = 3×1
      1.0883
      1.1042
      1.1041

E_ang = 3×1
      0.0388
      0.3424
      0.2216
```

(b) Unreduced prefault Y-bus

i. Without Load Shunt

main.mlx | Y1_prim_prim = 12×12 complex

	1	2	3	4	5	6	7	8	9	10	11	12
1	0.0000 -33.8085i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 +17.3611i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 +16.4474i	0.0000 + 0.0000i	0.0000 + 0.0000i
2	0.0000 + 0.0000i	0.0000 -24.3472i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 +16.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 8.3472i	0.0000 + 0.0000i
3	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 -22.5805i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 +17.0648i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 5.5157i
4	0.0000 +17.3611i	0.0000 + 0.0000i	0.0000 + 0.0000i	3.3075 -39.4758i	-1.9423 +10.5106i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	-1.3652 +11.6041i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i
5	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	-1.9423 +10.5106i	3.2242 -16.0988i	-1.2819 + 5.5882i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i
6	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 +17.0648i	0.0000 + 0.0000i	-1.2819 + 5.5882i	2.4365 -32.4373i	-1.1546 + 9.7843i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i
7	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	-1.1546 + 9.7843i	2.7710 -23.4824i	-1.6164 +13.6981i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i
8	0.0000 + 0.0000i	0.0000 +16.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	-1.6164 +13.6981i	2.8038 -35.6733i	-1.1874 + 5.9752i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i
9	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	-1.3652 +11.6041i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	-1.1874 + 5.9752i	2.5526 -17.5793i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i
10	0.0000 +16.4474i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 -16.4474i	0.0000 + 0.0000i	0.0000 + 0.0000i
11	0.0000 + 0.0000i	0.0000 + 8.3472i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 - 8.3472i	0.0000 + 0.0000i
12	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 5.5157i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 - 5.5157i

ii. With Load Shunt

main.mlx | Y1_prim = 12×12 complex

	1	2	3	4	5	6	7	8	9	10	11	12
1	0.0000 -33.8085i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 +17.3611i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 +16.4474i	0.0000 + 0.0000i	0.0000 + 0.0000i
2	0.0000 + 0.0000i	0.0000 -24.3472i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 +16.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 8.3472i	0.0000 + 0.0000i
3	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 -22.5805i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 +17.0648i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 5.5157i
4	0.0000 +17.3611i	0.0000 + 0.0000i	0.0000 + 0.0000i	3.3075 -39.4758i	-1.9423 +10.5106i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	-1.3652 +11.6041i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i
5	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	-1.9423 +10.5106i	4.1879 -16.4200i	-1.2819 + 5.5882i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i
6	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 +17.0648i	0.0000 + 0.0000i	-1.2819 + 5.5882i	2.4365 -32.4373i	-1.1546 + 9.7843i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i
7	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	-1.1546 + 9.7843i	3.8133 -23.8472i	-1.6164 +13.6981i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i
8	0.0000 + 0.0000i	0.0000 +16.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	-1.6164 +13.6981i	2.8038 -35.6733i	-1.1874 + 5.9752i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i
9	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	-1.3652 +11.6041i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	-1.1874 + 5.9752i	3.9350 -18.1323i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i
10	0.0000 +16.4474i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 -16.4474i	0.0000 + 0.0000i	0.0000 + 0.0000i
11	0.0000 + 0.0000i	0.0000 + 8.3472i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 - 8.3472i	0.0000 + 0.0000i
12	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 5.5157i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 + 0.0000i	0.0000 - 5.5157i

(c) Reduced prefault Y-bus

```
Y1 = 3×3 complex
     0.7938 - 3.2554i   0.2553 + 1.3546i   0.1826 + 1.0938i
     0.2553 + 1.3546i   0.3903 - 2.8662i   0.1902 + 0.9846i
     0.1826 + 1.0938i   0.1902 + 0.9846i   0.2541 - 2.4687i
```

(d) Reduced fault-on Y-bus

```
Y2 = 3×3 complex
     0.6640 - 3.9591i   0.1218 + 0.7214i   0.0000 + 0.0000i
     0.1218 + 0.7214i   0.2550 - 3.4355i   0.0000 + 0.0000i
     0.0000 + 0.0000i   0.0000 + 0.0000i   0.0000 - 4.1684i
```

(e) Reduced post-fault Y-bus

```
Y3 = 3×3 complex
     0.9961 - 2.6093i   0.2420 + 1.1768i   0.1081 + 0.5032i
     0.2420 + 1.1768i   0.3830 - 2.8198i   0.1740 + 1.1405i
     0.1081 + 0.5032i   0.1740 + 1.1405i   0.2267 - 1.9460i
```

(b) Discrete equation to solve the program based on trapezoidal integration scheme is shown below: With damping applied, based on swing equation

$$\frac{d\delta_i}{dt} = \omega_s \Delta\omega_i, \qquad \frac{d\Delta\omega_i}{dt} = \frac{1}{2H_i}(P_{mi} - P_{ei} - D_i \Delta\omega_i)$$

And:

$$P_{ei} = \sum_{j=1}^{n} E_i E_j Y_{ij} \cos(\theta_{ij} - \delta_i + \delta_j)$$

For trapezoidal rule, with a step size of h, we have :

$$x_{k+1} = x_k + \frac{h}{2}\left(f(x(t_k)) + f(x(t_{k+1}))\right)$$

We derive:

$$f_{\delta i}^{c} = \frac{1}{2}\left[\omega_s(\Delta\omega_i^k + \Delta\omega_i^{k+1})\right]$$

$$f_{\omega i}^{c} = \frac{1}{2}\left[\frac{1}{2H_i}(P_{mi} - P_{ei}^{k} - D_i\Delta\omega_i^k) + \frac{1}{2H_i}(P_{mi} - P_{ei}^{k+1} - D_i\Delta\omega_i^{k+1})\right]$$

Therefore:

$$\delta_i^{k+1} = \delta_i^k + \frac{h}{2}\left(\omega_s(\Delta\omega_i^k + \Delta\omega_i^{k+1})\right)$$

$$\Delta\omega_i^{k+1} = \Delta\omega_i^k + \frac{h}{2}\left[\frac{1}{2H_i}(2P_{mi} - P_{ei}^{k} - D_i\Delta\omega_{ik} - P_{ei}^{k+1} - D_i\Delta\omega_i^{k+1})\right]$$

For each generator i = 1,2,3, there should be two equations

$$F_{\delta i} = \delta_i^{k+1} - \delta_i^k - \frac{h}{2}\left(\omega_s(\Delta\omega_i^k + \Delta\omega_i^{k+1})\right) = 0$$

$$F_{\Delta\omega i} = \Delta\omega_i^{k+1} - \Delta\omega_i^k - \frac{h}{2}\left[\frac{1}{2H_i}(2P_{mi} - P_{ei}^{k} - D_i\Delta\omega_i^k - P_{ei}^{k+1} - D_i\Delta\omega_i^{k+1})\right] = 0$$

(c) Newton-Raphson iteration is used to iteratively solve the algebraic equation
Define:

$$F(X) = 0$$

where

$$X = \left[\delta_1^{k+1},\, \delta_2^{k+1},\, \delta_3^{k+1},\, \Delta\omega_1^{k+1},\, \Delta\omega_2^{k+1},\, \Delta\omega_3^{k+1}\right]^T$$

and

$$F(X) = [F_{\delta_1},\, F_{\delta_2},\, F_{\delta_3},\, F_{\Delta\omega_1},\, F_{\Delta\omega_2},\, F_{\Delta\omega_3}]^T$$

For newton

$$\text{At iteration } r, \qquad J^{(r)}\,\Delta X^{(r)} = -F\left(X^{(r)}\right)$$

$$\text{then update} \qquad X^{(r+1)} = X^{(r)} + \Delta X^{(r)}$$

The jacobian for this problem is in the block form of :

$$J = \begin{bmatrix} \frac{\partial F_\delta}{\partial \delta} & \frac{\partial F_\delta}{\partial \Delta\omega} \\ \frac{\partial F_{\Delta\omega}}{\partial \delta} & \frac{\partial F_{\Delta\omega}}{\partial \Delta\omega} \end{bmatrix}$$

The top left block:

$$\frac{\partial F_\delta}{\partial \delta} = I \text{ The identity matrix}$$

The top right block:

$$\frac{\partial F_\delta}{\partial \Delta\omega} = -\frac{h\omega_s}{2} I$$

The bottom left block

$$\frac{\partial F_{\Delta\omega}}{\partial \delta} = \frac{h}{4H}\frac{\partial P_{ei}}{\partial \delta_j}$$

For i = j which is the diagonal component :

$$\frac{\partial F_{\Delta\omega}}{\partial \delta} = \frac{h}{4H_i}\sum_{j\neq i}^{n} E_i E_j Y_{ij}\sin(\theta_{ij} - \delta_i + \delta_j) = A$$

For i $\neq$ j which is the non-diagonal component

$$\frac{\partial F_{\Delta\omega}}{\partial \delta} = -\frac{h}{4H_i}(E_i E_j Y_{ij} \sin(\theta_{ij} - \delta_i + \delta_j) = B$$

The bottom right block

$$\frac{\partial F_{\Delta\omega}}{\partial \Delta\omega} = (1 + \frac{hD_i}{4H_i})I = C$$

Therefore, the overall Jacobian matrix is:

$$J = \begin{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} & \begin{bmatrix} -\frac{h\omega_s}{2} & 0 & 0 \\ 0 & -\frac{h\omega_s}{2} & 0 \\ 0 & 0 & -\frac{h\omega_s}{2} \end{bmatrix} \\ \begin{bmatrix} A & B & B \\ B & A & B \\ B & B & A \end{bmatrix} & \begin{bmatrix} C & 0 & 0 \\ 0 & C & 0 \\ 0 & 0 & C \end{bmatrix} \end{bmatrix}$$

(d) The programme is extended to matlab to simulate the fault. The complete code is attached.

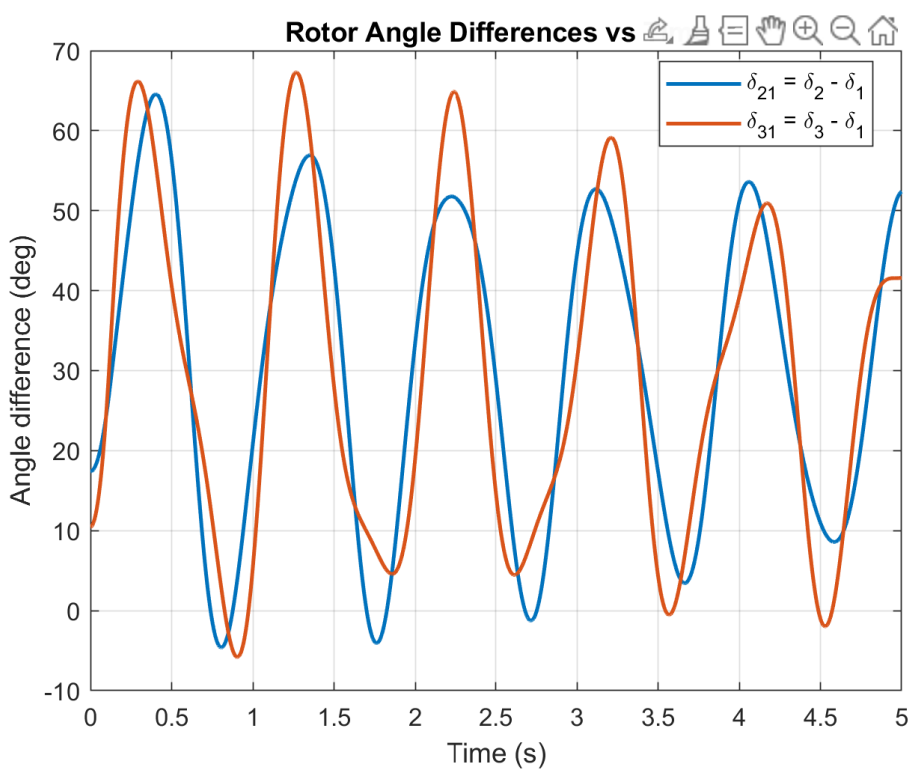

Figure 1: Relative Angle Difference for a critical clearing time of 0.12s

With a fault clearing time of 0.12s, the system remains transiently stable. The rotor angle show oscillations after the disturbance, which is expected following a disturbance at 0s. However, the oscillations stay bounded over the entire simulation period and do not show any trend of growing without limit. This indicates that synchronism is preserved after fault clearing.

(e) The critical clearing time is 0.2240s. For a fault clearing time of 0.2240s, the rotor angle differences remain bounded. The oscillations are very large, indicating the system is operating at the stability limit. But the generators still remain in synchronoism over the simulation window. When the clearing time is increased to 0.2241s, the response becomes unstable. Angle difference between generators diverges rapidly to infinity, showing loss of synchronism. The code to find the critical clearing time is attached.

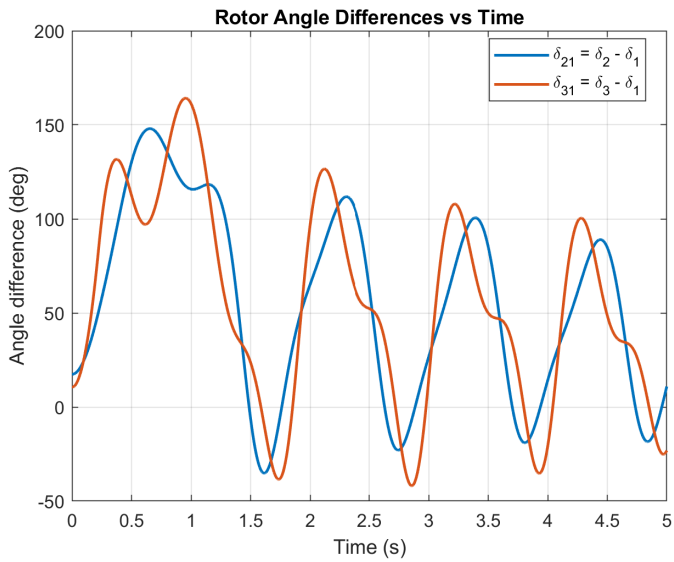

Figure 2: Relative Angle Difference for a critical clearing time of 0.224s

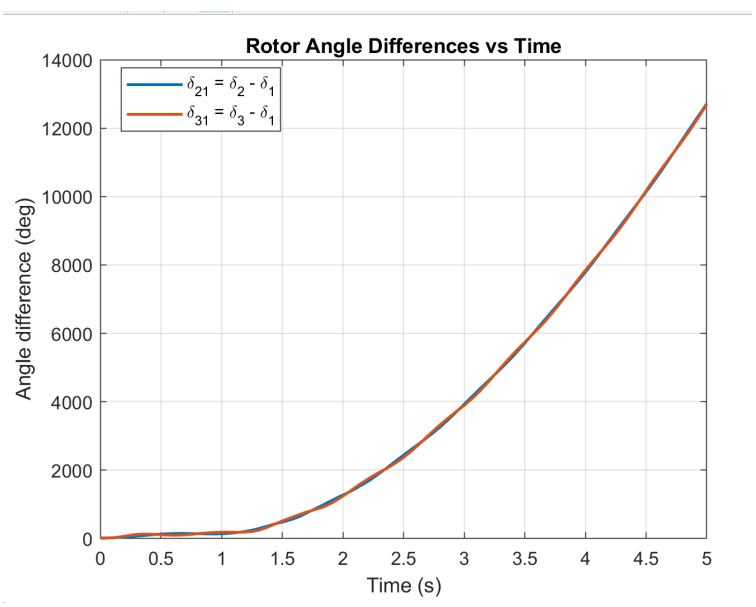

Figure 3: Relative Angle Difference for a critical clearing time of 0.2241s